Why Frontier Production Starts with One Golden Snapshot

Prove the assembled pipeline before multiplying it 161 times

GOTHAM PDF reconstruction notes

June 2026

Status: The argument for one golden run is strong. The run itself is still pending.

1 What we are trying to explain

The reducer has passed synthetic tests, GPU tests, real-shard tests, and a four-GPU/eight-shard comparison. Why not launch all 161 snapshots immediately?

Because the remaining failures would hit every snapshot the same way. If the HIP build, complete-snapshot checks, 1024-rank reduction, output identity, or MI250X performance is wrong, a production array does not give 161 independent experiments. It repeats the same mistake 161 times.

2 The simple picture

The parts have passed tests. The assembled production machine has not.

Run one exact, complete snapshot with archived controls. Make it exercise the whole assembled path and expose performance. Then start production at concurrency one, using the first larger 256-node snapshot as a second scale canary.

3 The cartoon version

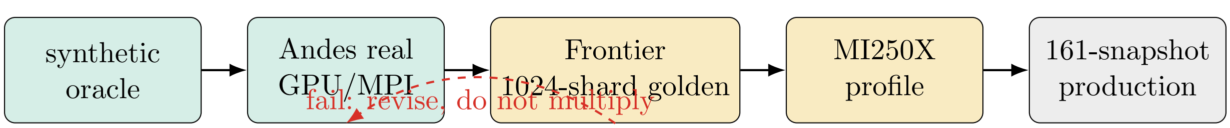

Figure 1: Each gate removes a failure mode that the previous gate cannot test. *This is a schematic, not evidence.*

The important arrow is the one pointing backward. A failed golden run sends us back to the tested Andes path; it does not fan out into production.

4 The more precise version

The golden case is not an arbitrary sample. It is the exact 8 pc snapshot with archived controls and a manageable 128-node footprint:

Simulation / phase	`res_8pc / phase2`
Source cube	`00028`
Archived PDF identity	`00062` at 10.520002963096713
Expected source cycle	202829
Source shards / Frontier ranks	1024
Frontier nodes	128
Expected domain volume	400^3 code-volume units

For archived control histogram A and rebuilt histogram R,

$$E_{\mathrm{L1}} = \frac{\sum_b |R_b - A_b|}{\sum_b |A_b|}, \qquad E_{\mathrm{total}} = \frac{|\sum_b R_b - \sum_b A_b|}{\sum_b |A_b|}.$$

The absolute-weight denominator is necessary because flux histograms are signed.

The implemented global AMR checks require contiguous shards, unique logical leaf keys, no ancestor/descendant pair, and the expected total volume. That is substantially stronger than a partial run. It is not yet a mathematical proof that every physical region is covered exactly once; geometry-to-key checks are still needed to exclude a compensating hole and overlap.

A job exit code is not enough. The golden run passes only if all of the following are true:

1. a frozen executable checksum, source revision, and product catalog;
2. exact contiguous shard manifest and matching complete header digests;
3. valid records, hydro state, unique AMR leaves, no ancestor overlap;
4. leaf-volume closure to 400^3 within relative tolerance 2×10^{-9};
5. exact output number and exact float64 archived PDF time;
6. publication accepted only when the final manifest and validator report both exist;
7. passing archived `output29` radial and `output31` controls, including $E_{\mathrm{L1}} < 2 \times 10^{-5}$ for the `output29` radial marginal and $E_{\mathrm{L1}} < 2 \times 10^{-3}$ for full `output31`;
8. repaired `output29` angle populated rather than copied from the corrupted archive;
9. predeclared acceptable `original`, `science`, and `all` throughput, memory, and walltime on MI250X.

That is the gate. Anything less is a partial success.

5 What this idea predicts

- The complete golden run should pass the declared archived-control limits. Similarity to the partial real-data errors is a useful diagnostic, not the release definition.

- It should disagree with the broken full `output29` while matching its radial marginal.
- It should activate and pass global checks deliberately skipped by partial runs.
- If the all-product path is atomic-bound, the original repair set may pass while science/all throughput requires optimization.
- A failed golden run should block every production array, even if the failure occurs after hours of otherwise plausible output. This is a proposed property; the current submitter does not enforce it.

6 What the evidence actually shows

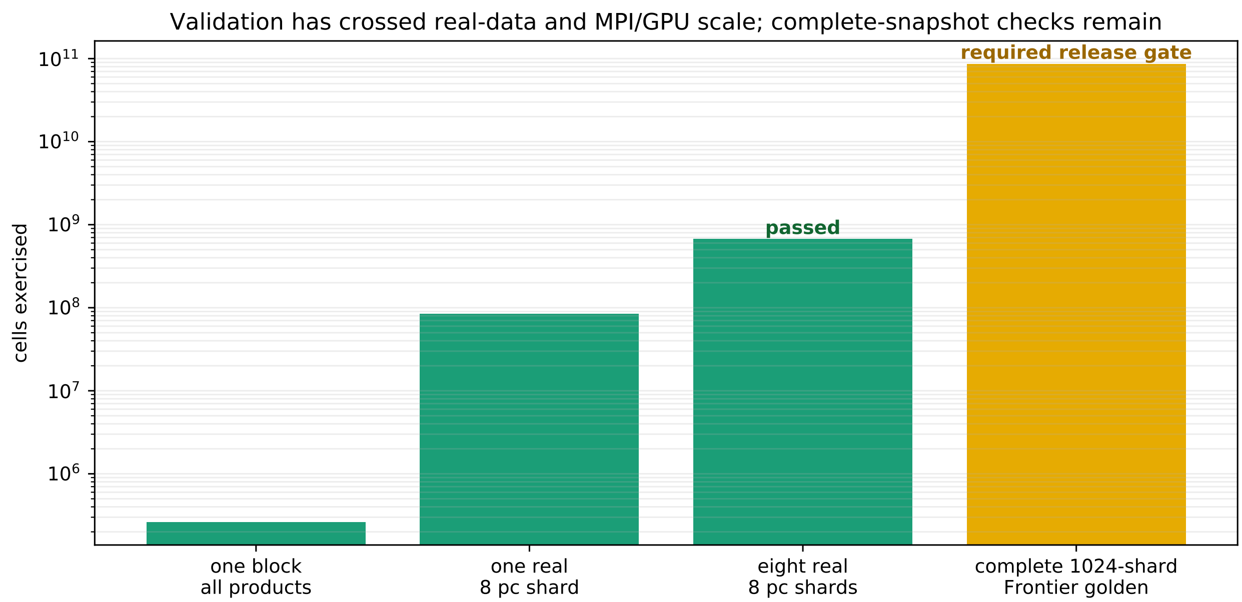

Figure 2: The validation ladder built from completed artifacts and the pending golden requirement. The final cell count is an exact fixed-record inventory from live source-file sizes, not a measured completed run.

What to look at. The green steps are finished. The first amber step crosses a new line: for the first time, the code can check the whole AMR volume against its implemented logical-key and volume invariants. The earlier steps test equations, format parsing, GPU execution, and a partial MPI reduction. They do not see the whole domain. The pending step contains exactly 86,467,149,824 cells by live fixed-record inventory; that is scale, not a completed result.

Caveat. The green steps tell us the equations and partial pipeline are behaving. They do not certify a complete production snapshot. The partial-run log explicitly says that global AMR closure is skipped for intentionally limited input.

What to look at. Look at the yellow part of the `res_8pc` bar. Those 21 snapshots lack archived PDF controls, so they depend more heavily on the internal invariants and on confidence earned from the controlled golden run. For those snapshots, there is no archived answer to rescue us after the fact. The pipeline has to earn trust before they run.

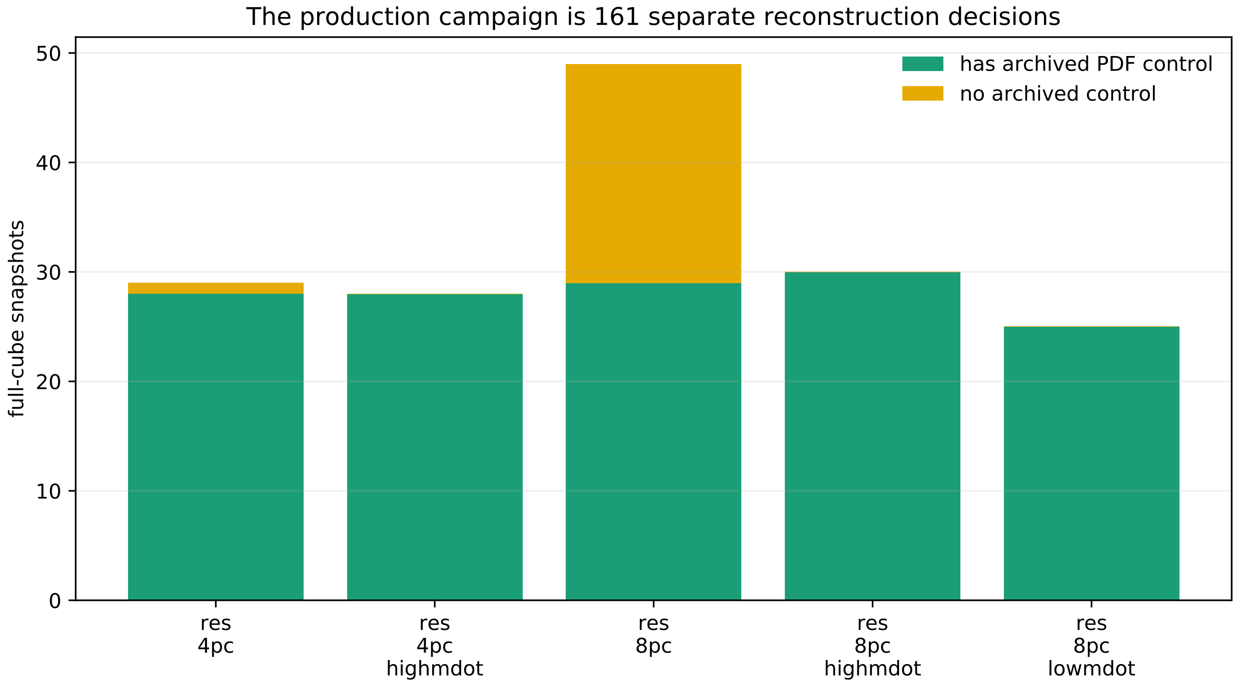

Figure 3: Static Frontier inventory. There are 161 full cubes. 140 have matching archived PDF identities and exact times; 21 uniform-phase snapshots do not.

7 The key figures

The ladder shows the missing test plainly: none of the completed steps sees the whole domain. The inventory shows the exposure being protected. The corruption document's control-error figure gives us reason to expect the golden controls to pass, but not permission to skip them.

8 What works

- The synthetic/reference and release-gate suite records six passing tests.
- One-block CPU/GPU agreement is near roundoff.
- One complete real shard and eight complete real shards have run.
- The four-GPU/eight-shard job completed with exit code `0:0`.
- The matched real-data controls pass by orders of magnitude.
- Static manifests map 140 source cubes to exact archived PDF identities.
- Jobs default to plan mode, refuse existing output, and require explicit confirmation tokens.

9 What does not work

Frontier HIP has not been built and run. No 1024-rank reduction has completed. No complete-snapshot AMR report exists. The all-product path has not been profiled on MI250X. Frontier filesystem behavior is unknown. Some 4 pc snapshots require 256 nodes, twice the golden node count. No numerical performance budget currently defines "acceptable," and the validator does not yet enforce every exact output-identity field described by the proposed gate.

More importantly, the current gate is advisory. The golden job runs only the lightweight manifest check and tells the operator to run `validate_real_8pc.py`; the production submitter does not require a passing golden JSON report or tie it to an executable checksum. Routine production jobs also do not execute the 140 mapped archived controls.

So the golden snapshot is necessary, but it is not permission for high concurrency. The first 256-node snapshot and at least one no-control uniform snapshot remain explicit canaries.

10 How this could be wrong

Maybe this is overkill. If the complete run taught us nothing that the eight-shard test had not already shown, then spending 128 nodes on a gate would be ceremony. But the complete run is the first test that can catch missing shards, duplicate logical leaves, ancestor/descendant pairs, a wrong total domain volume, failures in the 1024-rank reduction, and actual Frontier throughput. It cannot yet exclude every arbitrary physical hole or overlap.

The opposite risk is treating one golden success as universal proof. It is not. The five simulations share schema and code, but differ in size and phase inventory. Per-snapshot manifest validation and archived controls should remain required, but generic per-snapshot archived-control validation is not yet implemented.

11 What would decide the issue

Run the golden Frontier job with `PRODUCTS=original`. Before that run, freeze the exact source and executable checksum, make the job execute the real-snapshot validator, and make production submission require its passing immutable JSON report. Then benchmark `science` and `all` against predeclared resource limits and add geometry-to-key validation. Use `res_4pc`, phase `phase2a`, sequence `00003` as the first 256-node canary. Use `res_8pc`, phase `phase2_uniform`, sequence `00003` as the first no-control canary. Only then release production at concurrency one.

12 Bottom line

Bottom line. The golden run is not caution for its own sake. It targets the specific properties that remain untested and that would fail across the whole campaign at once. The strategy leads; the current scripts do not yet enforce it. Freeze the artifact, prove one whole snapshot, run the 256-node and no-control canaries, then multiply.

A Revision notes

This document was revised after solution-advocate, logic/technical, evidence/figure, red-team, visual-design, human-editor, and reproducibility checks. Pass two states that the current gate is advisory, requires a frozen artifact and immutable validator report, narrows the AMR guarantees, adds predeclared performance budgets, and promotes the 256-node and no-control cases to explicit canaries. Review records live in the local `notes/` directory.